Smart Public Transport Analytics
Business Insights

Teyzix Core - Data Analytics Internship - Task DA-3

Overview

This document presents the business insights derived from the analysis of 1,300 public-transport trips across eight routes over a four-month period. The operation carried 42,025 passengers and generated ₨3.75 million in ticket revenue, with an average delay of 11.1 minutes and an average occupancy of 73.4%. All figures are drawn from the cleaned dataset.

1. Highest Revenue Routes

Revenue is led by high-fare routes rather than the busiest ones:

Route	Revenue (₨)
R01 – City Center – Airport	725,496
R08 – South Terminal – Business District	565,020
R05 – West Gate – Tech Park	514,800

R01 earns the most because of its premium airport fare, not because it carries the most passengers. Revenue leadership is driven by pricing and route type, not raw volume.

2. Most Crowded Routes

Occupancy is high but remarkably even across routes (72–75%), so no single route stands out as dramatically overcrowded. The true crowding signal appears by **bus type and time**: Mini buses run fullest at 76.2% occupancy, and crowding concentrates in peak commuting hours rather than on specific routes.

3. Peak Passenger Hours

Ridership shows two sharp daily peaks: **7–08 AM and 5–06 PM**. The single busiest hour is 7 AM (5,140 passengers), followed by 5 PM (4,618) and 6 PM (4,563) — the classic morning and evening commuter rush.

4. Delay Hotspots

Delays are driven overwhelmingly by **weather and peak hours, not by route**. Average delay nearly doubles in poor weather — Fog (15.8 min), Storm (15.4), and Rain (15.4) versus Clear (9.0) and Cloudy (8.7). The worst hours mirror the passenger peaks (6 PM: 15.4 min, 7 AM: 14.9 min).

5. Low Performing Routes

The weakest routes by revenue are:

Route	Revenue (₨)
R06 – Old Town – Hospital	287,548
R02 – North Station – University	352,881
R07 – Riverside – Stadium	422,152

R06 earns only 40% of what R01 earns. These are low-fare routes; their weakness is revenue, not necessarily ridership — R07 has among the highest average passengers per trip despite mid-low revenue.

6. Operational Inefficiencies

8.3% of all trips are cancelled (108 trips), a meaningful loss of service, clustering on routes R05 (10.8%) and R08 (9.3%). Weekend average ridership (28.5 per trip) is notably lower than weekday (33.7), suggesting weekend service may be over-supplied relative to demand.

7. Resource Utilization

Fleet utilisation is healthy at 73.4% average occupancy, leaving limited spare capacity at peak. Fuel efficiency varies sharply by bus type: Electric buses consume just 0.03 litres per passenger versus 0.57 for Mini and 0.32 for AC — Electric is by far the most resource-efficient.

8. Passenger Demand Trends

Demand is stable month to month (9,500–11,300 passengers), dipping slightly in February, and is **time-driven rather than seasonal**: weekday commuter peaks dominate and weekend demand is consistently lower. This predictability is an operational advantage for planning.